

AWOID MAJOR UNPACKING - KEEP BOXES PACKED

Maintenance sheet version 10 for ablation area (tripod, one level)

Station name	TAS_A_v3
People present	CHAD, CLAN + ASD
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	9/9-24 + 11/9-24

Notes: Install a new station, v3_rewired

- NOT TRANSMITTING
- CONSIDER BUILDING IN ADVANCE

DRILL: {1HR}
THERMISTOR
PTA
SR50 STAKE

• FWD AMJIS PHONE?
• EXPECTED 1m/y MELT

AWS to do list		Check after maintenance			
Part	Shipped (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer	Y (244327)	Y			Clean? Y / N
Ablation hose assembly	Y (50209)	Y			No air in hose and bladder half full? Y / N
Iridium modem IMEI number	Y (inside loggerbox)	Y			3005 3406 4381 670
Inclinometer/compass	Y (inside CNR4)	Y	Attached to the radiometer? Y / N		
Satellite antenna	Y	Y	Not relevant		
Wind sensor	Y	Y	Black box pointing toward the radiometer (i.e., along the boom)? Y / N		
Temperature / humidity assembly	Y	Y	Fan spins? Y / N Sounds, OK? Y / N		
HygroClip T/hum assembly	N	Y	NB: Rotronics (Y / N) could have a 40 °C offset. New or old type? NB: Lufft sensor (Y / N) Cabel not included, so not installed		
Sonic ranger on AWS	Y	Y	Old sensor with new membrane? Y / N		
Sonic ranger on stake(s)	Y		Old sensor with new membrane? Y / N NB: make sure not being buried next winter		
Thermistor string	Y (digital)	Y			
Solar panel	Y	Y	Clean? Y / N NB: is not effective when partially covered		
GPS antenna	Y (inside loggerbox)	Y	At top of logger enclosure? Y / N		
Logger box ID	T3B-14		CR1000 or CR1000X? ID number: AT3B-14		
Multiplexer	N		Clicking during measurement cycle? Y / N		
Logger enclosure	Y	Y	Replaced (including everything inside)? Y / N New desiccant bags? Y / N		
Battery box	Y	Y	New box? Y / N New desiccant bags? Y / N		
Stakes	Y		New one(s) drilled in? Y / N Length of new stake(s) 8 m? Y / N Bottom plugged to reduce melt-in? Y / N		
Tripod	Y	Y	New tripod? Y / N New sensor boom? Y / N		

Station name	TAS_A_v3
Date	

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)		— NA
Download data and/or change CF/SD Card (wait for green light)	Y / N	Y / N
Name of logger program		pram v3 - Remind - L5.c
Screen dump of values in fast scan mode	Y / N	Y / N
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y / N	Y / N
Mast tilt towards radiometer (- if tilting towards anemometer)	°	2 °
Mast tilt across boom looking from radiometer (- if clockwise)	°	1 °
Boom direction relative to north	° true/magn.	° true/magn.
Box on wind sensor exactly towards mast? If not, measure.	Y / N	Y / N
Radiometer exactly aligned with inclinometer? If not, measure.	Y / N	Y / N
Temperature / humidity height (bottom of casing – surface)	cm	NA cm
Stake assembly standing?	Y / N / sort of	Y / N / sort of
Free length of stake (if 3: outer, holding boom with sonic ranger)	cm	163 cm
Free length of stake 2 (if 3: middle)	cm	— cm
Free length of stake 3 (if 3: outer, without sonic ranger)	cm	— cm
Height of ablation bladder fluid level above the ice surface	cm	160 cm
Vertical difference in surface height (at station – at hose) (approx.)	cm	~0 cm
Length of thermistor string on surface from surface marking	cm	665 cm
Apparent damage to sensors? If yes, which?	Y / N	Y / N

Notes: lufft cable was missing → thus not installed

* Huge crevasse North of station

* 7 drill flights stuck in ice.

Station re-visited on Sep. 11 2024 to install lufft sensor. Station was not re-measured after tilting.

Must-do at next visit:

* Install lufft ⇒ DONE!

Maintenance sheet version 10 for ablation area (tripod, one level)

Station name	TAS_A_SF
People present	
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	

Notes: Install new SF and power setup

NO BATTERY, MIGHT BE MISSING
STATION

ENSURE PERFECT V3 STATION
BEFORE SF MAINTENANCE

AWS to do list		Check after maintenance			
Part	Shipped (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer	X N				Clean? Y / N
Ablation hose assembly	X N				No air in hose and bladder half full? Y / N
Iridium modem IMEI number	Reuse from TAS_A_v2 or old TAS_A_SF				3002 3406 6316 280
Inclinometer/compass	N		Attached to the radiometer? Y / N		
Satellite antenna	Reuse from TAS_A_v2 or old TAS_A_SF				
Wind sensor	N		Black box pointing toward the radiometer (i.e., along the boom)? Y / N		
Temperature / humidity assembly	N		Fan spins? Y / N Sounds, OK? Y / N		
HygroClip T/hum assembly	N		NB: Rotronics (Y / N) could have a 40 °C offset. New or old type? NB: Lufft sensor (Y / N)		
Sonic ranger on AWS	Y		Old sensor with new membrane? Y / N		
Sonic ranger on stake(s)	N		Old sensor with new membrane? Y / N NB: make sure not being buried next winter		
Thermistor string	N				
Solar panel	Reuse from TAS_A_v2 or old TAS_A_SF		Clean? Y / N NB: is not effective when partially covered		
GPS antenna	Inside loggerbox		At top of logger enclosure? Y / N		
Logger box ID	T1SF-0001		CR1000 or CR1000X? ID number: 1013		
Multiplexer	N		Clicking during measurement cycle? Y / N		
Logger enclosure	N		Replaced (including everything inside)? Y / N New desiccant bags? Y / N		
Battery box	Y		New box? Y / N New desiccant bags? Y / N		
Stakes	N		New one(s) drilled in? Y / N Length of new stake(s) 8 m? Y / N Bottom plugged to reduce melt-in? Y / N		
Tripod	N		New tripod? Y / N New sensor boom? Y / N		

Station name	TAS_A_SF
Date	

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)		
Download data and/or change CF/SD Card (wait for green light)	Y / N	Y / N
Name of logger program		
Screen dump of values in fast scan mode	Y / N	Y / N
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y / N	Y / N
Mast tilt towards radiometer (- if tilting towards anemometer)	°	°
Mast tilt across boom looking from radiometer (- if clockwise)	°	°
Boom direction relative to north	° true/magn.	° true/magn.
Box on wind sensor exactly towards mast? If not, measure.	Y / N	Y / N
Radiometer exactly aligned with inclinometer? If not, measure.	Y / N	Y / N
Temperature / humidity height (bottom of casing – surface)	cm	cm
Stake assembly standing?	Y / N / sort of	Y / N / sort of
Free length of stake (if 3: outer, holding boom with sonic ranger)	cm	cm
Free length of stake 2 (if 3: middle)	cm	cm
Free length of stake 3 (if 3: outer, without sonic ranger)	cm	cm
Height of ablation bladder fluid level above the ice surface	cm	cm
Vertical difference in surface height (at station – at hose) (approx.)	cm	cm
Length of thermistor string on surface from surface marking	cm	cm
Apparent damage to sensors? If yes, which?	Y / N	Y / N

Notes:

Must-do at next visit:

Maintenance sheet version 10 for ablation area (tripod, one level)

Station name	TAS_L
People present	CLAN, CHAO + ASO
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	11/9 -24

Notes: REDRILL SR50 STAKES
 . CHANGE TASP, WCL, SR50
 . FROM 810

AWS to do list		Check after maintenance			
Part	Shipped (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer	N				Clean? ☒ Y / N
Ablation hose assembly	N				No air in hose and bladder half full? Y / N, Not a lot of fluid
Iridium modem IMEI number	N				3002 3401 0884 770
Inclinometer/compass	Y	Y			Attached to the radiometer? Y / ☒ N
Satellite antenna	N				
Wind sensor	N				Black box pointing toward the radiometer (i.e., along the boom)? ☒ Y / N
Temperature / humidity assembly	Y	Y			Fan spins? ☒ Y / N Sounds, OK? ☒ Y / N
HygroClip T/hum assembly	N				NB: Rotronics (Y / N) could have a 40 °C offset. New or old type? NB: Lufft sensor (Y / N)
Sonic ranger on AWS	Y	Y			Old sensor with new membrane? Y / ☒ N
Sonic ranger on stake(s)	Y	Y			Old sensor with new membrane? Y / ☒ N NB: make sure not being buried next winter
Thermistor string	N				
Solar panel	N				Clean? ☒ Y / N NB: is not effective when partially covered
GPS antenna	N				At top of logger enclosure? ☒ Y / N
Logger box ID	N				CR1000 or <u>CR1000X</u> ? ID number:
Multiplexer	N				Clicking during measurement cycle? ☒ Y / N
Logger enclosure	N				Replaced (including everything inside)? Y / N New desiccant bags? Y / N
Battery box	N				New box? Y / ☒ N New desiccant bags? Y / N
Stakes	Y				New one(s) drilled in? ☒ Y / N Length of new stake(s) 8 m? ☒ Y / N Bottom plugged to reduce melt-in? Y / ☒ N
Tripod	N				New tripod? Y / ☒ N New sensor boom? ☒ Y / N

Station name	TAS_L
Date	11/9/2024

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)	+03:07 Screen shot	
Download data and/or change CF/SD Card (wait for green light)	Y/N	Y/N
Name of logger program	PROMICE V2.2022 Screen shot V1.2	
Screen dump of values in fast scan mode	Y/N	Y/N
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y/N	Y/N
Mast tilt towards radiometer (- if tilting towards anemometer) $+4^\circ$	82.5°	$+1^\circ$
Mast tilt across boom looking from radiometer (- if clockwise) $+2^\circ$	0°	-1°
Boom direction relative to north XX	90° true/magn.	$\sim 0^\circ$ true/magn.
Box on wind sensor exactly towards mast? If not, measure.	Y/N	Y/N
Radiometer exactly aligned with inclinometer? If not, measure.	Y/N	Y/N
Temperature / humidity height (bottom of casing – surface)	271 cm	267 cm
Stake assembly standing?	Y/N / sort of	Y/N / sort of
Free length of stake (if 3: outer, holding boom with sonic ranger)	0 cm	~ 150 cm
Free length of stake 2 (if 3: middle)	0 cm	— cm
Free length of stake 3 (if 3: outer, without sonic ranger)	0 cm	— cm
Height of ablation bladder fluid level above the ice surface	138.5 cm	182 cm
Vertical difference in surface height (at station – at hose) (approx.)	2 cm	+20 cm
Length of thermistor string on surface from surface marking	898.5 cm	898.5 cm
Apparent damage to sensors? If yes, which? XXX	Y/N	Y/N

Notes: *SR50 on stakes melted out + cable out.

*PTA: Genind på pose ikke skrevet ordentlig på, så har lakkert væske

*Hele tripod var veltet → boom bøjet

□) Langden er fra overfladen af hul til 1m markeringen

XX) station has been laying down; and raised up

XXX) +-String Cable damage (see photo), lapper kablet med self-vulcanisering tape

Must-do at next visit:

- Radiometer meget beskadiget på sensorer ⇒ cleaned and re-used
- Defekt kort – så det er fjernet. ⇒ ingen CF-kort i logger.

~~Plugs~~ Plugs on loggerbox are fragile

New station recommended for next visit.

Maintenance sheet version 10 for ablation area (tripod, one level)

Station name	TAS_A_v2
People present	CHA0, CLAN + A50
Purpose	Maintenance / installation / <u>take-down</u> / quick visit / emergency
Date	9/9-24

Notes: Take down station completely

SUNSET

AWS to do list		Check after maintenance			
Part	Shipped (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer	N				Clean? Y / N
Ablation hose assembly	N				No air in hose and bladder half full Y / N
Iridium modem IMEI number	N				
Inclinometer/compass	N		Attached to the radiometer? Y / N		
Satellite antenna	N				
Wind sensor	N		Black box pointing toward the radiometer (i.e., along the boom)? Y / N		
Temperature / humidity assembly	N		Fan spins? Y / N Sounds, OK? Y / N		
HygroClip T/hum assembly	N		NB: Rotronics (Y / N) could have a 40 °C offset. New or old type? NB: Lufft sensor (Y / N)		
Sonic ranger on AWS	N		Old sensor with new membrane? Y / N		
Sonic ranger on stake(s)	N		Old sensor with new membrane? Y / N NB: make sure not being buried next winter		
Thermistor string	N				
Solar panel	N		Clean? Y / N NB: is not effective when partially covered		
GPS antenna	N		At top of logger enclosure? Y / N		
Logger box ID	N		CR1000 or CR1000X? ID number:		
Multiplexer	N		Clicking during measurement cycle? Y / N		
Logger enclosure	N		Replaced (including everything inside)? Y / N New desiccant bags? Y / N		
Battery box	N		New box? Y / N New desiccant bags? Y / N		
Stakes	N		New one(s) drilled in? Y / N Length of new stake(s) 8 m? Y / N Bottom plugged to reduce melt-in? Y / N		
Tripod	N		New tripod? Y / N New sensor boom? Y / N		

Station name	TAS_A_v2
Date	

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)		
Download data and/or change CF/SD Card (wait for green light)	Y / N	Y / N
Name of logger program		
Screen dump of values in fast scan mode	Y / N	Y / N
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y / N	Y / N
Mast tilt towards radiometer (- if tilting towards anemometer)	°	°
Mast tilt across boom looking from radiometer (- if clockwise)	°	°
Boom direction relative to north	° true/magn.	° true/magn.
Box on wind sensor exactly towards mast? If not, measure.	Y / N	Y / N
Radiometer exactly aligned with inclinometer? If not, measure.	Y / N	Y / N
Temperature / humidity height (bottom of casing – surface)	cm	cm
Stake assembly standing?	Y / N / sort of	Y / N / sort of
Free length of stake (if 3: outer, holding boom with sonic ranger)	cm	cm
Free length of stake 2 (if 3: middle)	cm	cm
Free length of stake 3 (if 3: outer, without sonic ranger)	cm	cm
Height of ablation bladder fluid level above the ice surface	cm	cm
Vertical difference in surface height (at station – at hose) (approx.)	cm	cm
Length of thermistor string on surface from surface marking	cm	cm
Apparent damage to sensors? If yes, which?	Y / N	Y / N

Notes:

Station was tipped over, instruments frozen in ice
 Station not transmitting
 Disassembled and shipped home

Must-do at next visit:

7/-